

## Multiple Choice (10 marks)

1. The maximum number of acute angles in a convex 10-gon in the Euclidean plane is
  - (a) 1
  - (b) 2
  - (c) 3
  - (d) 4
2. Statement 1 | Every solvable group is of prime-power order. Statement 2 | Every group of prime-power order is solvable.
  - (a) True, True
  - (b) False, False
  - (c) True, False
  - (d) False, True
3. Find the degree for the given field extension  $\mathbb{Q}(\sqrt{2})$  over  $\mathbb{Q}$ .
  - (a) 0
  - (b) 2
  - (c) 1
  - (d) 3
4. Find the maximum possible order for an element of  $S_n$  for  $n = 7$ .
  - (a) 6
  - (b) 12
  - (c) 30
  - (d) 105
5. Statement 1 | If  $a$  and  $b$  are elements of finite order in an Abelian group, then  $|ab|$  is the lcm  $(|a|, |b|)$ . Statement 2 | If  $g$  is a group element and  $g^n = e$ , then  $|g| = n$ .
  - (a) True, True
  - (b) False, False
  - (c) True, False
  - (d) False, True

## True / False (10 marks)

*Answer True or False*

6. True or False: The product of (2,3) and (3,5) in the ring  $\mathbb{Z}_5 \times \mathbb{Z}_9$  is (1,6).
7. True or False: The circle defined by  $x^2 + y^2 = 16$  has more points of intersection with the parabola  $x^2 = y + 4$  than any other circle listed.
8. True or False: The area of an equilateral triangle with an inscribed circle radius of 2 is 16 square units.
9. True or False: In the group  $G = \{2, 4, 6, 8\}$  under multiplication modulo 10, the identity element is 2.
10. True or False: The maximum possible order for an element of the symmetric group  $S_{10}$  is 12.

## Long Form Response (20 marks)

*Answer each question with a clear and complete explanation.*

11. The inhabitants of the island of Jumble use the standard Kobish alphabet (20 letters, A through T). Each word in their language is 4 letters or less, and for some reason, they insist that all words contain the letter A at least once. How many words are possible?
12. Discuss the implications of the statements regarding subgroups in group theory: Analyze why the product of subgroups  $HK$  is a subgroup if one of them is normal, and assess the properties of groups of order  $p^2$  being Abelian.

*End of Paper*